

The doctrine of temporal parts

February 12th, 2004

Different versions of the doctrine

- In the previous lecture, we stated a version of the doctrine of the temporal parts as follows:
 - Whenever an object exists at a time, the object has a temporal part at that time.
 - x is a temporal part of y at t means: x exists only at t , and x is part of y , and x shares a part with every part of y that exists at t .

- This definition of ‘temporal part’ uses a notion of *parthood* that is *not temporally relativised*.
 - Some opponents of the doctrine of temporal parts claim not to understand this notion: they will reject the previous statement of the view as *meaningless*.
 - But there are other versions of the view that do not use this notion, and are (plausibly) consequences of this statement.

- First alternative: a purely spatiotemporal doctrine.
 - Whenever an object x exists at a time t , there is an object y that exists *only* at t , and occupies the same region of space as x at t .

- Second alternative: a doctrine stated in terms of a *temporally relativised* notion of parthood.
 - Whenever an object x exists at a time t , there is an object y that exists only at t , [and is part of x at t], and is such that for every z that is part of x at t , z and y share some part at t . (Sider, p. 99)
 - two things are said to *overlap* [at a time] if they share some part [at that time].
- Third alternative: like the second, but leaving out the clause in brackets.

- These doctrines all concern the existence of *instantaneous* temporal parts — ‘timeslices’. A closely-related view concerns the existence of temporal *segments* of objects:
 - Whenever an object exists throughout a certain period of time, it has a temporal segment that exists throughout that period and at no other times.
 - How is ‘temporal segment’ to be defined? Here we face the same range of options as before.

Temporal parts and unrestricted composition

- The doctrine of temporal parts must be distinguished from the view that *nothing lasts for more than an instant*.
- Is the doctrine of temporal parts *consistent* with this view?
 - Yes. If you look at the definitions, and bear in mind that everything counts as an ‘improper’ part of itself, you’ll see that an instantaneous object is its own temporal part at the time at which it exists.

- Things look different when someone, like Sider, combines the doctrine of temporal parts with the **principle of unrestricted composition**.
 - According to this principle, whenever there are some things—no matter how distantly they might be related, in space or in time—there is an object which is composed of them, and hence has all of them as parts.
 - Other ways to say ‘it is composed of them’ which you might come across: ‘it is their fusion’; ‘it is their mereological sum’.

- Unrestricted composition lets us ‘make’ new things by ‘summing’ the timeslices of other things.
- It follows from the combination of these views that if any material object occupies a region of space at a time, then *lots* of material objects occupy that region at that time. (Assuming that there are lots of times at which material objects exist).
 - The fusion of any thing which occupies that region of space at that time with any thing or things which don’t exist at that time will be another thing that occupies that region of space at that time.

Temporal parts and the puzzles of material constitution.

- If you hold the combination of these views, it is very easy to explain our justification for believing the sentences which led to the original paradoxes of material constitution.
- You *already* think, for example, that there is an object which starts to exist when the clay is shaped into a statue, and ceases to exist when it is squashed. The only remaining question is how we are justified in believing that this thing is a *statue*. But this seems to be just a matter of understanding the concept of a statue. If our concept had worked differently, it would have picked out an object that traced out a different spatiotemporal region.

- Given the doctrines of temporal parts and unrestricted composition, the only questions that remain to be answered about any of the puzzle cases are questions about which, among the many material objects that are present in any situation, are the ones picked out by our concepts and names.

- Compare what Sider says about the Ship of Theseus puzzle (p. 12):

Four-dimensionalism does not, on its own, answer our question, but the *metaphysical* puzzle has been dissolved. We have a perfectly clear metaphysical picture of what happens: the world contains space-time worms corresponding to both answers to our question. The only remaining question is the merely *conceptual* one of which of these spacetime worms counts a ship.

The doctrine of temporal parts and change

- Opponents of the view sometimes claim that if the doctrine of temporal parts is true, nothing ever *changes*. (The ‘block universe’)
- But this is false: believers in temporal parts generally accept the following analysis of change:

‘x changes [between t and t’]’ means that x has dissimilar temporal parts [at t and t’].
- The real controversy should be over whether this is an *adequate* analysis.

Temporally relativised predication

- Sider suggests the following view:

For x to have long hair at t is for x 's temporal part at t to have long hair [simpliciter].

- A natural view for a proponent of temporal parts to adopt.

- One might be tempted to generalise this:
‘ x is F at t ’ means that x ’s temporal part at t is F .
- Will this work?
- Consider: I am, right now, a human being and college professor.
 - Surely we don’t want to say that my current temporal part is a human being or a college professor.
 - Or the sum of it with some past dinosaur.